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## Malaysia: Reserve Requirements, AFC

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# Malaysia: Reserve Requirements, AFC<sup>1</sup>

*Bailey Decker<sup>2</sup>*

Yale Program on Financial Stability Case Study  
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## **Abstract**

Bank Negara Malaysia (BNM) unpegged the ringgit in July 1997, days after Thailand floated the baht. Ringgit depreciation and adverse investor sentiment worsened, contributing to a domestic liquidity shortage and capital flight. Malaysia experienced market instability in the early months of 1998, particularly pressure on its exchange rate, foreign currency reserves, and interest rates. At the same time, disruptions in the domestic money market and loan intermediation process caused an increase in lending rates, which resulted in debt servicing problems and weakened financial stability. To facilitate lending and productive economic activity, BNM twice lowered the statutory reserve requirement (SRR) at commercial banks, merchant banks, and finance companies on February 16 and July 1, 1998. BNM simultaneously offset the additional liquidity added to the banking system by reducing direct central bank lending to the interbank market and raised the three-month intervention rate (three-month interbank rate) to better reflect liquidity conditions in the market. When inflationary pressure subsided toward the end of summer 1998, BNM began to pursue monetary easing. Thus, the central bank did not sterilize the SRR reductions on September 1 and 16, 1998. BNM paired these latter, unsterilized SRR reductions with a lower three-month interbank rate to promote liquidity and enhance banking institutions' lending capacity. The initial SRR reductions would have released MYR 22 billion (USD 5.8 billion) into the banking system, had BNM not lowered direct lending to the interbank market. The latter SRR reductions released MYR 15 billion into the banking system.

**Keywords:** Asian Financial Crisis, liquidity rules, reserve ratio, reserve requirements, Malaysia

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<sup>1</sup> This case study is part of the Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering adjustment of reserve requirements. Cases are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/>.

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## Overview

In the early 1990s, the Malaysian economy enjoyed relatively strong fundamentals: The real GDP growth rate averaged 8.5% per year, unemployment fell below 3%, exchange rates were stable, and foreign liabilities did not exceed Malaysia's substantial buildup of international reserves (BNM 1998k; BNM 1999; Kaplan and Rodrik 2001; Meesook et al. 2001; Sundaram 2006). However, the floating of the Thai baht on July 2, 1997, triggered the Asian Financial Crisis (AFC), exposing vulnerabilities in the Malaysian economy (BNM 1998k; Feldstein 1998; Meesook et al. 2001; Poon 1999). In the years preceding the AFC, Malaysian authorities' investment-led growth strategy depended on readily reversible portfolio investments of uncertain quality (Kaplan and Rodrik 2001; Meesook et al. 2001; Poon 1999; Sundaram 2006). After defending the ringgit's value,<sup>3</sup> Bank Negara Malaysia relented and unpegged the ringgit on July 14, 1997 (Meesook et al. 2001; Poon 1999). Malaysia felt the acute effects of the crisis later than other countries in the region. Conditions continued to worsen even after pressure eased in Thailand and the Republic of Korea (Korea) by the beginning of 1998 (see Figure 1) (BNM 1999; Kaplan and Rodrik 2001). Adverse investor sentiment and ringgit depreciation contributed to a domestic liquidity shortage and capital outflows estimated at more than USD 10 billion from early 1997 to mid-1998 (Meesook et al. 2001).

Unlike Thailand, Indonesia, and Korea, Malaysia did not seek a bailout from the International Monetary Fund (IMF) (AP-Dow Jones News Service 1997; Poon 1999; Sundaram 2006). Prime Minister Mahathir Mohamad cited economic and political reasons in an address on December 31, 1997: "[The IMF] will

## Key Terms

Purpose of Adjusting Reserve Requirement (RR): During monetary tightening, to "enhance the efficiency of the intermediation process and not to provide additional liquidity to the system" (BNM 1998a); during monetary easing, to "ease liquidity in the banking system" (BNM 1998h)

|                                        |                                  |
|----------------------------------------|----------------------------------|
| Range of RR Ratio (RRR) Peak-to-Trough | 13.5%–4% of eligible liabilities |
|----------------------------------------|----------------------------------|

|                     |                                           |
|---------------------|-------------------------------------------|
| RRR Increase Period | Pre-crisis (set at 13.5% on June 1, 1996) |
|---------------------|-------------------------------------------|

|                     |                                |
|---------------------|--------------------------------|
| RRR Decrease Period | February 1998 – September 1998 |
|---------------------|--------------------------------|

|                 |                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------|
| Legal Authority | Central Bank of Malaysia Act 1958 (Revised—1994) and Banking and Financial Institutions Act 1989 |
|-----------------|--------------------------------------------------------------------------------------------------|

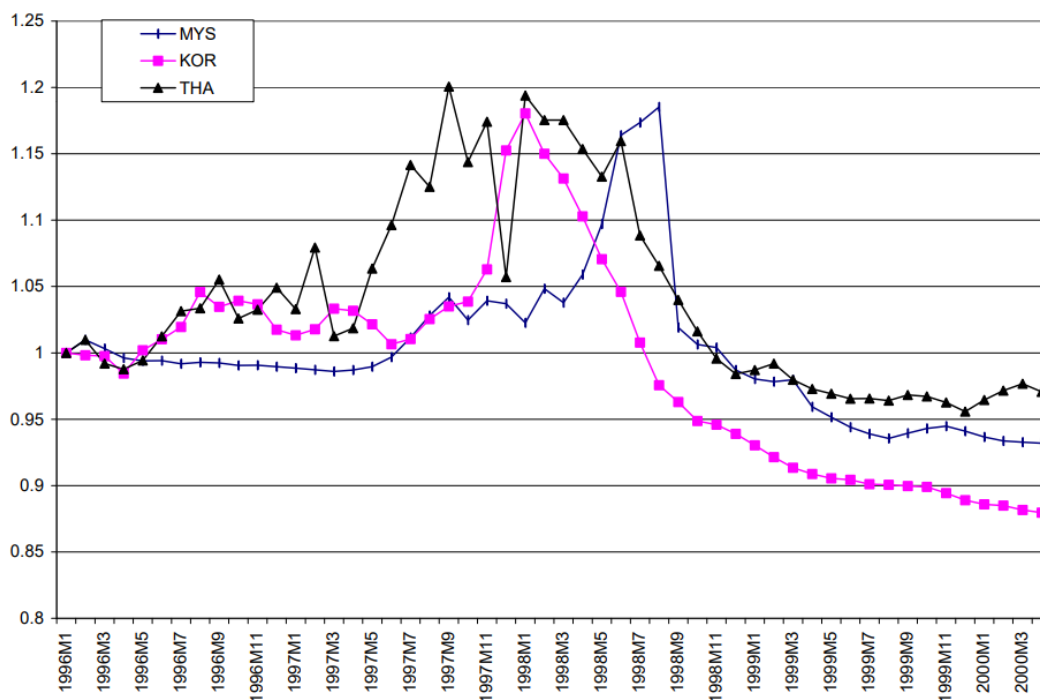
|                                   |               |
|-----------------------------------|---------------|
| Interest/Remuneration on Reserves | Unremunerated |
|-----------------------------------|---------------|

|                  |                                                                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notable Features | BNM first cut the SRR during monetary tightening but cut the SRR further as it switched to monetary easing<br><br>Malaysia concurrently replaced a longstanding liquid asset requirement with a liquidity management framework |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|          |                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Outcomes | MYR 22 billion (USD 5.8 billion) injected into the banking system with the initial cuts, offset by decreased direct lending to the interbank market; MYR 15 billion injected with the later cuts |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

<sup>3</sup> These operations included intervention in the exchange market and an elevated interbank rate. BNM briefly raised the interbank rate to 35%, then lowered it to 10% (Meesook et al. 2001).

force us to increase taxes, raise our interest rates, close a large number of our banks and finance companies and let companies which are weak because of the decline in the value of the ringgit go bankrupt” (AP-Dow Jones News Service 1997). In its 1998 Annual Report, BNM argued that “the orthodox International Monetary Fund (IMF) policy prescription of tight fiscal and monetary policies would only have destabilized the situation further and induced a much sharper contraction in real output” (BNM 1999, 10). According to BNM, international investors perceived it as a sign of weakness that Malaysia did not participate in an IMF program, because “international investors did not differentiate the risks in countries in the Asian region” (BNM 1999, 14). In fact, BNM said, Malaysia would not have qualified for an IMF program, because of its strong balance of payments and relatively limited foreign debt (BNM 1999).

**Figure 1: Financial Pressure Market Index (Jan. 1996=1)**

Note: On the left axis, Kaplan and Rodrik (2019) calculated the financial market pressure index as “a weighted average of the (log) exchange rate, (log) foreign currency reserves (with declines in reserves contributing positively to the index), and the interest rate” (19).

Source: Kaplan and Rodrik 2001.

In retrospect, Malaysia’s response to the AFC fell into four phases. First, Prime Minister Mahathir led a series of defensive policies from July to December 1997 (Sundaram 2006). In this initial phase, BNM cut interest rates and the government tried to raise a loan in international capital markets, but failed after a credit downgrade (Poon 1999). Second, from December 1997 to summer 1998, Deputy Prime Minister and Finance Minister Anwar Ibrahim tightened fiscal and monetary policy to contain inflationary pressure, while taking steps to promote financial stability (Kaplan and Rodrik 2001; Sundaram 2006). Measures during this period included cuts to government spending by 18%, intervention in the foreign exchange interbank market, full deposit guarantees of all financial institutions, and more restrictions on stockbroking companies (Meesook et al. 2001; Sundaram 2006). Third, Anwar led a reflationary period from summer 1998 until his dismissal from the cabinet and later arrest in September, 1998. This included an attempt to temper the downturn through spending policies. Fourth, Mahathir introduced exchange and capital controls to insulate monetary policy and other domestic government interventions from volatile foreign markets (Meesook et al. 2001; Sundaram 2006). This included repegging the ringgit at MYR 3.80 per

US dollar,<sup>4</sup> restricting the repatriation of portfolio capital, and prohibiting offshore ringgit activities (BNM 1999; Kaplan and Rodrik 2001; Meesook et al. 2001; Sundaram 2006).

The focus of this case study—lowering the statutory reserve requirement (SRR)—began during the monetary tightening phase and continued into the monetary easing phase (Sundaram 2006). During this second period of tightened fiscal and monetary policy, illiquidity in domestic financial markets disrupted corporate borrowers' access to credit. BNM lowered the SRR to encourage commercial banks, merchant banks, and finance companies to finance productive economic activity (BNM 1998a; BNM 1999). For the initial SRR cuts on February 16 and July 1, 1998, BNM announced that it would reduce direct lending on the interbank market to sterilize any liquidity impacts of the SRR easing and contain base money growth (BNM 1998a; BNM 1998e). BNM also raised the three-month intervention rate (henceforth, three-month interbank rate) to maintain tight monetary policy (BNM 1998a). However, by August, BNM had shifted to an expansionary monetary policy stance, so it did not sterilize the later SRR cuts on September 1 and 16, 1998 (BNM 1998h; BNM 1998j). BNM also lowered the three-month interbank rate (BNM 1998h). At the same time, BNM changed its method for calculating required reserves. It widened the band for permissible daily variation in the average balances required to meet the SRR, giving banking institutions more flexibility in managing their liquidity (BNM 1998d).

### Summary Evaluation

Banks' deposits with BNM declined by MYR 88.5 billion (USD 23 billion)<sup>5</sup> in 1998, according to its 1998 Annual Report (BNM 1999). Required statutory reserves declined by MYR 42 billion (BNM n.d.b).

BNM lowered the SRR to facilitate the intermediation process and stoke productive economic activity. Overall, total loans issued by the banking system declined in 1998 by MYR 7.6 billion or 1.8% due to economic contraction, high interest rates, higher repayments in response to higher interest rates, and the stalling of expansion plans. This downward trend occurred from February to September 1998.<sup>6</sup> However, the central bank saw an "encouraging" turnaround in October and November 1998, as issued loans began to increase. The central bank noted in its 1998 Annual Report that loan approvals amounted to MYR 24 billion in the first half of the year and MYR 38.2 billion in the second half of the year (BNM 1999).

BNM attributed the turnaround in banking activity in 1998 to its policy measures. It provided data describing the impact of its reductions to the SRR in the two phases in 1998 (BNM 1999).

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<sup>4</sup> The exchange rate hovered around USD 1.00 = MYR 2.50 in the years leading up to the crisis (Sundaram 2006).

<sup>5</sup> USD 1 = MYR 3.80 on December 31, 1998, according to the IMF.

<sup>6</sup> BNM noted decreased lending "for consumption credit, purchase of securities, purchase of transport vehicles and community, social and personal services and manufacturing sectors" during this time (BNM 1999).

First, the central bank estimated that the initial SRR cuts on February 16 and July 1, 1998, would have released MYR 22 billion into the banking system if it were not for the sterilizing effect of its reduced direct lending on the interbank market (BNM 1998a; BNM 1998e). BNM also attributed declining money market rates and a subsequent reduction in lending rates during this period to its SRR cuts. Just in the month of February 1998, lending rates fell by 5.2%, from 21.5% to 16.3% (BNM 1999).

Later, BNM estimated that the SRR reductions on September 1 and 16, 1998, released MYR 15 billion into the banking system (BNM n.d.b). The central bank also said that monetary easing during this phase—including unsterilized reductions to the SRR and reductions to the three-month interbank rate—led to declines in the average base lending rates (BLRs) of commercial banks and finance companies. The average BLR of commercial banks fell from 11.7% in August to 8.0% in November 1998. The average BLR of finance companies fell from 14.2% in August to 9.5% in November 1998 (BNM 1999).



| Context: Malaysia 1997–1998                                                                                 |      |                             |
|-------------------------------------------------------------------------------------------------------------|------|-----------------------------|
| GDP (SAAR, nominal GDP in LCU converted to USD)                                                             |      | \$100.0 billion in 1997     |
|                                                                                                             |      | \$72.2 billion in 1998      |
| GDP per capita (SAAR, nominal GDP in LCU converted to USD)                                                  |      | \$4,638 in 1997             |
|                                                                                                             |      | \$3,263 in 1998             |
| Sovereign credit rating (five-year senior debt)                                                             | 1997 | Moody's: Data not available |
|                                                                                                             |      | S&P: AA                     |
|                                                                                                             |      | Fitch: Data not available   |
|                                                                                                             | 1998 | Moody's: A3                 |
|                                                                                                             |      | S&P: A-                     |
|                                                                                                             |      | Fitch: BBB                  |
| Size of banking system                                                                                      |      | \$162.4 billion in 1997     |
|                                                                                                             |      | \$117.8 billion in 1998     |
| Size of banking system as a % of GDP                                                                        |      | 162.3% in 1997              |
|                                                                                                             |      | 163.3% in 1998              |
| Size of banking system as a % of financial system                                                           |      | Data not available in 1997  |
|                                                                                                             |      | Data not available in 1998  |
| Five-bank concentration of banking system                                                                   |      | Data not available in 1997  |
|                                                                                                             |      | Data not available in 1998  |
| Foreign involvement in banking system                                                                       |      | Data not available in 1997  |
|                                                                                                             |      | Data not available in 1998  |
| Existence of deposit insurance                                                                              |      | No in 1997                  |
|                                                                                                             |      | No in 1998                  |
| Sources: Bloomberg; World Bank Global Financial Development Database; World Bank Deposit Insurance Dataset. |      |                             |

## Key Design Decisions

- 1. Purpose: BNM sterilized the initial SRR cuts to promote lending without interfering with its tight monetary policy. After BNM began monetary easing in August 1998, it did not sterilize the later SRR cuts, thus promoting liquidity.**

By early 1998, illiquidity in domestic financial markets disrupted corporate borrowers' access to credit. Lending rates rose above the official target interbank rate, creating debt servicing problems in the corporate sector. In response, BNM lowered the SRR for commercial banks, merchant banks, and finance companies (BNM 1998a; BNM 1999). The central bank said in press releases that it would sterilize the SRR cuts on February 16 and July 1, 1998, to maintain its tight monetary policy (BNM 1998a; BNM 1998e).

As inflationary pressure subsided, BNM began monetary easing in August 1998 (BNM 1998h). Thus, BNM did not sterilize the SRR cuts on September 1 and 16, 1998 (BNM 1998h; BNM 1998j). These SRR reductions along with the reduced interbank rate would ease liquidity in the banking system so that borrowers could more readily obtain loans at a lower rate (BNM 1998h; BNM 1998j; BNM 1999).

- 2. Part of a Package: In the first half of 1998, BNM paired SRR reductions with other interventions to neutralize additional liquidity in the banking system, maintain tight monetary policy, and help individual banking institutions in their daily liquidity operations. As inflationary pressure eased, BNM also began "cautious" monetary easing.**

BNM sought to stem inflation from the ringgit's depreciation during the first half of 1998 (BNM 1998a; BNM 1999). To sterilize the additional liquidity that would have been released by the SRR reductions, BNM reduced direct lending on the interbank market in conjunction with its SRR reductions on February 16, 1998, and July 1, 1998 (BNM 1998a; BNM 1998b; BNM 1998e; BNM 1998f). BNM estimated that the February 16 and July 1 reductions would have released MYR 14 billion and MYR 8 billion of additional funds, respectively, if the central bank had not reduced direct lending (BNM 1998a; BNM 1998e).

To maintain tight monetary policy, BNM raised the three-month interbank rate from 10% to 11% with the first SRR reduction on February 16, 1998 (BNM 1998a).

BNM began "cautious" monetary easing in August 1998 and lowered the three-month interbank rate to 9.5% with the SRR reduction on September 1, 1998 (BNM 1998g).

Earlier in March 1998, BNM said that it would review the role of the the SRR and liquid asset requirement (LAR) to "fully [restore] investor confidence and [create] an environment that would support the return of the economy to a stronger growth path" (BNM 1998c). The LAR required that commercial banks hold at least 17% of their eligible liabilities in liquid assets; the requirement was 10% for finance companies and merchant banks (or 12.5% if they granted negotiable instruments of deposit) (BNM 1998g).

On July 31, 1998, BNM announced a new liquidity framework that phased out the LAR. Instead, the BNM required banking institutions to forecast their assets, liabilities, and off-balance-sheet commitments to meet near-term obligations in addition to “unexpected heavy withdrawal for at least one month” (BNM 1998g). In practice, commercial banks had to be able to withstand the withdrawal of 5% of their deposit base over a week or 7% over a month; merchant banks and finance companies had to be able to withstand the withdrawal of 5% of their deposit base over a week or 15% over a month (BNM n.d.a). BNM phased in the new framework over six months. It required companies to continue to comply with the LAR standard during the transition (BNM 1998g).

**3. Legal Authority: BNM used the Central Bank of Malaysia Act 1958 (Revised—1994) and the Banking and Financial Institutions Act 1989 to lower the SRR.**

The Central Bank of Malaysia Act 1958 (Revised—1994) established BNM as Malaysia’s central bank (CBM Act 1994, secs. 3–4). Under this framework, and with the finance minister’s approval, BNM could require banking institutions<sup>7</sup> to hold reserves in Malaysian or foreign currency, possibly differentiated by the size or location of banking institutions (CBM Act 1994, secs. 37, 39).

**4. Administration: BNM administered changes to the SRR with approval from the finance minister.**

With the finance minister’s approval, BNM required banking institutions to meet the SRR (CBM Act 1994, secs. 37, 39).

**5. Governance: The finance minister had some control over BNM’s board of directors.**

A board of directors was responsible for BNM’s policy and general administration (CBM Act 1994, sec. 8). The board included the governor, three or fewer deputy governors, and five to eight directors (CBM Act 1994, sec. 8). The board informed the finance minister of its monetary and banking policy. The finance minister could issue binding directives if he or she disagreed with their plans and operations. In the event that the board objected to the finance minister’s override, the House of Representatives would hear the objections and reasoning of both sides (CBM Act 1994, sec. 34).

Beyond legal frameworks, research did not uncover specific instances of oversight pertaining to SRR reductions during the AFC.

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<sup>7</sup> Under Section 2 of the Central Bank of Malaysia Act 1958 (Revised—1994), “banking institution” was an inclusive term that meant “a licensed bank, a licensed merchant bank or a licensed finance company as defined in the Banking and Financial Institutions Act 1989 or an Islamic Bank” (CBM Act 1994, sec. 2). Banking institutions obtained licenses by submitting applications to BNM and the finance minister with detailed memoranda on constituent documents, governance, audited balance sheets, and more (Banking and Financial Institutions Act 1989, sec. 5).

6. **Communication: BNM stated that the initial SRR cuts would be neutralized by decreased central bank lending to the interbank market. BNM did not sterilize the later SRR cuts and said that these cuts would inject liquidity into the banking system.**

BNM announced changes to the SRR through press releases<sup>8</sup> (BNM 1998a; BNM 1998e; BNM 1998h; BNM 1998j).

In first half of 1998, BNM sought tight monetary conditions (BNM 1998a). However, a contemporaneous Morgan Stanley report suggested that BNM's SRR reduction on February 16, 1998, had an easing effect. *The Business Times* quoted this report and characterization on March 9, 1998. BNM Deputy Governor Fong Weng Phak issued a follow-up press release in response on the same day. He pointed to reduced direct lending to the interbank market by BNM and higher interbank lending rates as evidence that monetary conditions remained tight (BNM 1998b). BNM issued a similar clarification press release on June 26, 1998, to reiterate its larger commitment to tightened monetary policy (BNM 1998f).

When the government began to pursue monetary easing in August 1998, BNM said that the "combined impact of the reduction in SRR and [three-month interbank rate would] ease liquidity in the system, making available more loanable funds to borrowers, and at a lower rate" (BNM 1998h; BNM 1998j).

7. **Assets Qualifying as Reserves: Banking institutions were required to hold their reserves in accounts at Bank Negara Malaysia.**

Banking institutions had to hold reserves at the central bank, whether in Malaysian ringgit or other foreign currencies (CBM Act 1994, sec. 37).

8. **Reservable Liabilities: Deposits and other short-term liabilities fell under the SRR with some exceptions.**

Research did not identify contemporaneous BNM public statements describing the liabilities covered by reserve requirements.

According to more recent BNM documentation, reservable liabilities included demand deposits, savings deposits, fixed deposits, specific investment deposits, general investment deposits, commodity murabahah,<sup>9</sup> call deposits, short-term deposits, investments linked to derivatives, other deposits, notional interest deductions (NIDs), repurchase agreements, special deposits (withholding tax), housing development account deposits, amounts due to designated financial institutions, miscellaneous borrowings, interest payable to

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<sup>8</sup> BNM's Communication Centre accepted queries about its policy measures by telephone from 8:30 am to 12:00am Malaysian Standard Time. International parties could send queries beyond these business hours by fax (BNM 1998i).

<sup>9</sup> Also known as cost-plus financing; an Islamic financing structure where the buyer and seller agree on an asset's cost and markup. The markup takes the place of interest, since interest is illegal under Islamic law. Thus, murabahah is a form of credit sale under Islamic law rather than an interest-bearing loan (Young 2020).

nonresidents, bills payable to nonresidents, and other miscellaneous liabilities due to nonresidents (BNM 2020).

Reservable liabilities excluded subordinated debt capital, exempt subordinated debt capital, subordinated debt capital with special approval from BNM, and recourse obligations on loans to Cagamas, the National Mortgage Corporation of Malaysia (BNM 2020).

#### **9. Computation: BNM widened the permissible daily variation in average balances required to meet the SRR.**

BNM computed the SRR for all banking institutions on the basis of the average amount of reserves each institution held during a two-week reserve maintenance period. BNM allowed an institution's reserves to vary over that period. In May 1998, BNM widened the band for permissible daily variation in reserve assets from  $\pm 0.5\%$  to  $\pm 2.0\%$ ,<sup>10</sup> to increase transparency and liquidity flows in the interbank market (BNM 1998d; BNM 1999).

#### **10. Eligible Institutions: BNM applied SRR changes to commercial banks, merchant banks, and finance companies.**

In its press releases, BNM stated that the SRR reductions applied to “commercial banks, merchant banks and finance companies” (BNM 1998a). BNM interchangeably used the inclusive term, banking institutions,<sup>11</sup> to refer to eligible commercial banks, merchant banks, and finance companies in communications to the public (BNM 1998a).

#### **11. Timing: BNM lowered the SRR during the second, third, and fourth phases of the crisis response.**

Malaysia began to acutely feel the effects of the AFC in the early months of 1998—after the crisis had peaked in Thailand and Korea (BNM 1999; Kaplan and Rodrik 2001). BNM lowered the SRR twice during the second phase of the government's crisis response: from 13.5% to 10% on February 16, 1998, and from 10% to 8% on July 1, 1998. The central bank reduced its direct lending to the interbank market to offset additional liquidity and set the three-month interbank rate at 11% to maintain tight monetary policy (BNM 1998a; BNM 1998e). By August 1998, BNM began easing monetary policy, as ringgit rates stabilized and inflation calmed (BNM 1998h). To this end, BNM also began to lower the three-month interbank rate (BNM 1998i). BNM reduced the SRR during the third phase of the crisis response on

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<sup>10</sup> In a press release announcing the computation change, BNM stated that the 2.0% band was based on a 20% variation on the prescribed SRR rate (BNM 1998d). It is unclear what the mechanics are behind the 20% variation.

<sup>11</sup> In its 1998 Annual Report, BNM defined the “banking system” similarly to its definition of “banking institution”: The banking system consisted of commercial banks, merchant banks, and finance companies (BNM 1999). According to the IMF, Malaysia had 37 commercial banks in 1994, of which 21 were domestic-owned, and 16 were foreign-owned. Commercial banks made up approximately 85% of the banking system's total assets in 1994 (IMF 1994). By year-end 1996, commercial banks accounted for 67% of total banking system assets (IMF 1997). According to data from the World Bank, total assets of the five largest banks as a share of total commercial banking assets were 61.6% in 1998 (World Bank n.d.).

September 1, 1998, from 8% to 6% (BNM 1998h). The central bank lowered the SRR during the fourth phase of crisis response on September 16, 1998, from 6% to 4% (BNM 1998j).

## **12. Changes in Reserve Requirements: BNM lowered the SRR four times between February and September 1998.**

Effective February 16, 1998, BNM announced a reduction in the SRR of commercial banks, merchant banks, and finance companies from 13.5% to 10% of eligible liabilities. The announcement stated that BNM would sterilize additional liquidity to the banking system by lowering its direct interbank lending, so as not to accelerate base money growth (BNM 1998a).

BNM began to publish information on April 30, 1998, about its daily operations—cash flow projections, liquidity operations, and money market tender results—to improve transparency (BNM 1998d).

On May 1, 1998, BNM widened the band for permissible daily variation in the average balances required to meet the SRR from  $\pm 0.5\%$  to  $\pm 2.0\%$ , to give banking institutions more flexibility in their daily liquidity operations (BNM 1998d; BNM 1999).

On July 1, 1998, BNM further reduced the SRR from 10% to 8% of eligible liabilities. This 2% reduction, they posited, would lower the BLR of commercial banks and finance companies by 0.21% and 0.27%, respectively (BNM 1998e). BNM anticipated an additional MYR 8 billion to the banking system, sterilized by reducing its interbank lending, and closely monitored interbank market liquidity (BNM 1998e; BNM 1998f).

Effective September 1, 1998, BNM dropped the SRR of eligible liabilities of commercial banks, merchant banks, and finance companies from 8% to 6% (BNM 1998h).

On September 16, 1998, BNM lowered the SRR from 6% to 4% of eligible liabilities at banking institutions, finance companies, and merchant banks to “improve the distribution of liquidity among individual banking institutions, thereby enhancing lending capacity of banking institutions” (BNM 1998j).

The two September cuts would inject about MYR 15 billion<sup>12</sup> into the banking system (BNM n.d.b).

See Figure 2 for a timeline of SRR reductions.

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<sup>12</sup> 4% of eligible liabilities (8% minus 4%) as of August 31, 1998.

**Figure 2: Timeline of SRR Reductions**

| Date               | Press Release Title                                     | Action                                                                                                                     |
|--------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| February 16, 1998  | "Monetary Policy"                                       | BNM lowered the SRR for commercial banks, merchant banks, and finance companies from 13.5% to 10% of eligible liabilities. |
| May 1, 1998        | "Measures to Enhance Liquidity Management"              | BNM widened the daily band for balances from $\pm 0.5\%$ to $\pm 2.0\%$ of the SRR rate averaged over a 14-day period.     |
| July 1, 1998       | "Statutory Reserve Requirement of Banking Institutions" | BNM lowered the SRR from 10% to 8% of eligible liabilities.                                                                |
| September 1, 1998  | "Easing of Monetary Policy"                             | BNM lowered the SRR from 8% to 6% of eligible liabilities.                                                                 |
| September 16, 1998 | "Statutory Reserve Requirement of Banking Institutions" | BNM lowered the SRR from 6% to 4% of eligible liabilities.                                                                 |

Sources: BNM 1998a; BNM 1998d; BNM 1998e; BNM 1998h; BNM 1998j.

### **13. Changes in Interest/Remuneration: The SRR was unremunerated.**

Malaysia's SRR was unremunerated, according to an October 1997 IMF report. Malaysian officials told the IMF that "the high level of unremunerated reserves impose[d] a significant implicit tax burden on financial intermediation" (IMF 1997, 15).

### **14. Other Restrictions: Research did not find additional restrictions for institutions eligible for SRR reductions.**

Beyond banking institutions—defined as licensed banks, licensed merchant banks, and licensed finance companies—research did not find additional restrictions for institutions eligible for reductions in the SRR (CBM Act 1994, sec. 2).

### **15. Impact on Monetary Policy Transmission: In the first half of 1998, BNM sterilized the liquidity created by the initial SRR cuts to avoid interfering with its tight-money policy. BNM did not sterilize the later SRR cuts, therefore injecting additional liquidity into the banking system in September 1998.**

BNM estimated that the February 16 and June 26 reductions to the SRR released MYR 14 billion and MYR 8 billion into the banking system, respectively, although it simultaneously offset these increases by cutting its direct lending on the interbank market (BNM 1998b; BNM 1998e).

By the end of July 1998, BNM determined that its overall policy package had successfully combated inflation; it said that “cautious” easing was in order (BNM 1998g).

BNM reported that the unsterilized SRR reductions on September 1 and 16 injected about MYR 15 billion into the banking system (BNM n.d.b).

**16. Duration: BNM did not announce a predetermined end to the SRR reduction.**

BNM announced the last change to the SRR from 6% to 4%, effective September 16, 1998. None of the press releases contained a predetermined end date to the SRR reduction<sup>13</sup> (BNM 1998j).

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<sup>13</sup> BNM maintained the SRR at 4% until November 24, 2008, when the central bank lowered the SRR from 4% to 3.5% during the Global Financial Crisis (BNM 2008; BNM n.d.c).



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